



Chennai Mathematical Institute

ADA Final Examination Report

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Applied Data Analysis (ADA)

Under the guidance of
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1 Problem 1: Economic Mobility Prediction

1.1 Introduction

This problem investigates whether communities with shorter average commute times exhibit higher upward economic mobility. The dataset covers 729 U.S. communities with:

- **Mobility** — Probability that a child born in 1980–82 to the lowest income quintile reaches the top quintile by age 30.
- **Commute** — Fraction of workers with commute < 15 minutes.
- **Latitude, Longitude** — Geographic location.
- **Name, State** — City and state information.
- **Latitude, Longitude** — Geographic coordinates of the community center.

We analyze univariate and bivariate patterns, perform linear regression, check assumptions, and fit a robust GLM.

1.2 Dataset Overview

The dataset contains:

- 729 communities
- 6 variables, no missing values
- 4 numeric variables: Mobility, Commute, Latitude, Longitude
- 2 categorical variables: Name, State

1.3 Summary Statistics

Variable	Mean	SD	Min	25%	Median	Max
Mobility	0.100	0.053	0.022	0.066	0.090	0.470
Commute	0.454	0.140	0.156	0.344	0.430	0.928
Longitude	-95.29	14.92	-170.72	-101.24	-93.46	-67.61
Latitude	38.94	6.03	19.58	34.76	38.83	68.37

Table 1: Summary Statistics

1.4 Exploratory Data Analysis

1.4.1 Distribution of Mobility

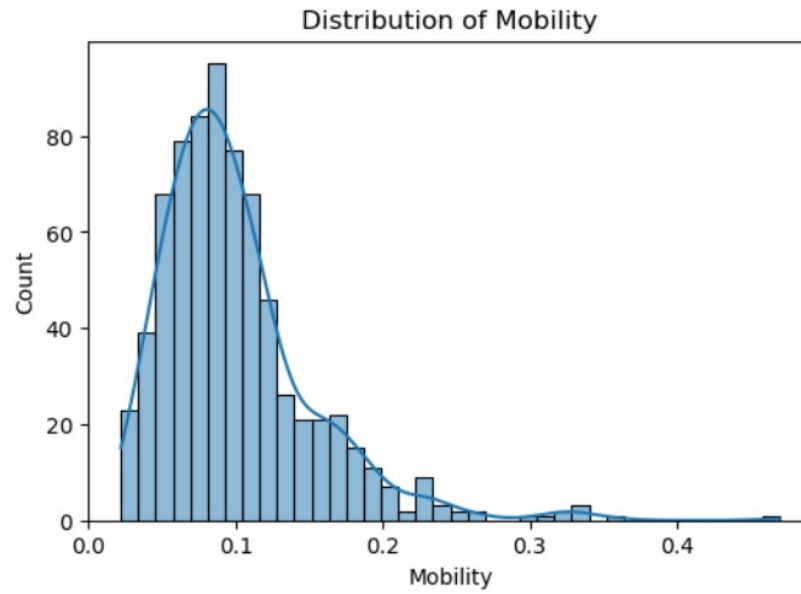


Figure 1: Distribution of Mobility

Mobility is right-skewed: a few communities exhibit very high mobility, while most lie near the lower end.

1.4.2 Distribution of Commute

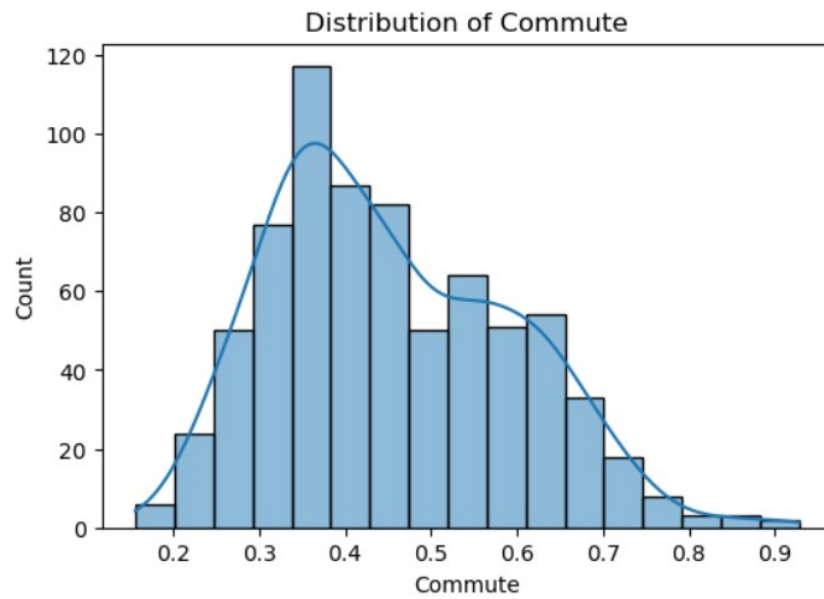


Figure 2: Distribution of Commute Fraction

Commute values are tightly clustered with few outliers, indicating consistency.

1.4.3 Correlation Analysis

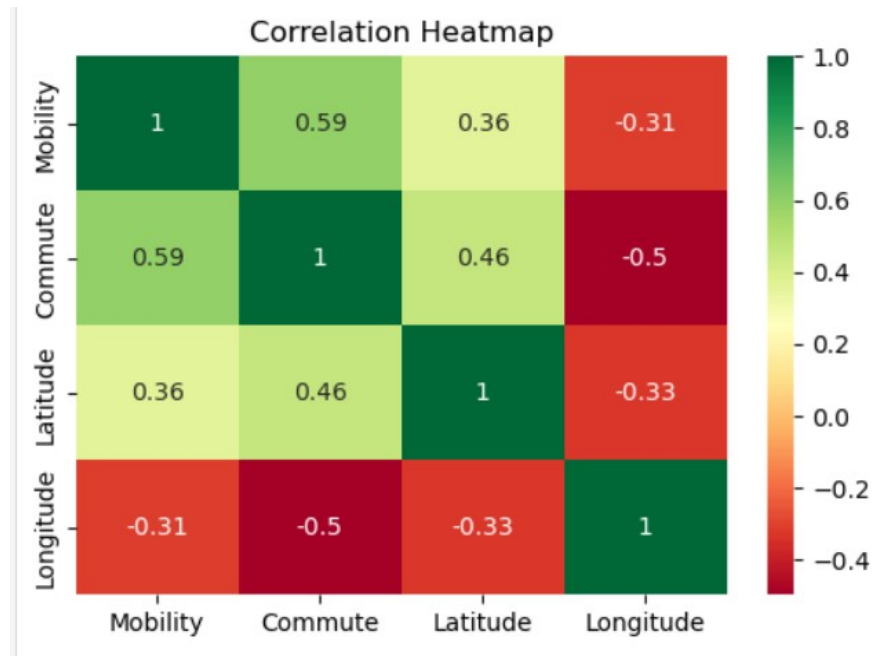


Figure 3: Correlation Heatmap

Key correlations:

- Mobility vs. Commute: ≈ 0.59 (positive, moderately strong)
- Commute vs. Longitude: ≈ -0.50

1.4.4 Geographical Mobility Patterns

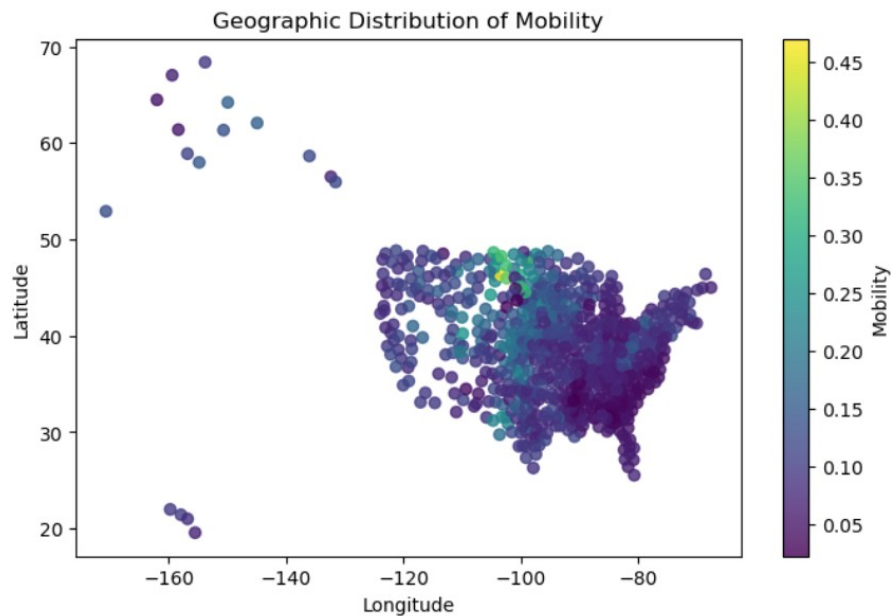


Figure 4: Geographic Mobility Distribution

Northern and Midwestern states show higher mobility than Southern states.

1.5 Model Formulation

We first fit a simple linear model:

$$\text{Mobility} = \beta_0 + \beta_1 \text{Commute} + \epsilon$$

Then a geographic model:

$$\text{Mobility} = \beta_0 + \beta_1 \text{Commute} + \beta_2 \text{Longitude} + \beta_3 \text{Latitude} + \epsilon$$

Residual diagnostics revealed violations of homoscedasticity and normality, leading to the use of a GLM with HC3 robust standard errors.

1.6 Model Diagnostics

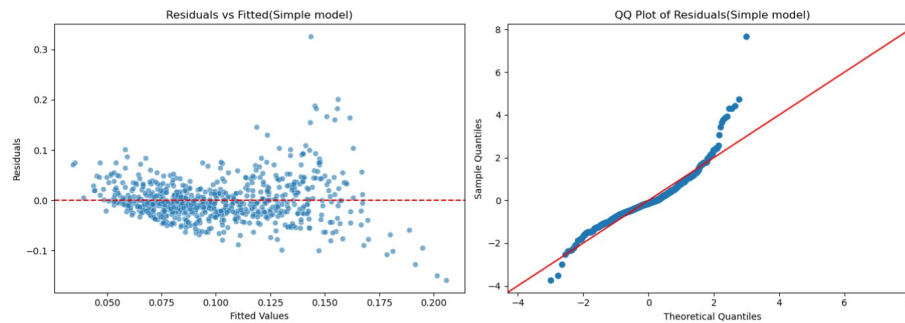


Figure 5: Residuals vs Fitted Values

Issues detected:

- Strong heteroscedasticity (BP test $p < 10^{-14}$)
- Residual autocorrelation (DW ≈ 1.34)
- Right-tail deviation in QQ plots

1.7 Estimation and Inference

1.7.1 Robust GLM: Simple Model

Predictor	Estimate	Robust SE	p-value
Intercept	-0.0002	0.007	0.977
Commute	0.2219	0.017	< 0.001

Table 2: HC3 Robust GLM (Simple Model)

Interpretation: A 0.1 increase in short-commute fraction predicts a 0.022 increase in mobility.

1.7.2 Robust GLM: Full Model

Predictor	Estimate	Robust SE	p-value
Commute	0.2010	0.016	< 0.001
Longitude	-0.00004	—	n.s.
Latitude	0.0009	0.0003	0.008

Table 3: HC3 Robust GLM (Geographic Model)

Commute remains the strongest predictor.

1.8 Predictive Models

We compare GLM vs. Random Forest:

Model	RMSE	MAE	R^2
GLM	0.044	0.03	0.4117
Random Forest	0.034	0.021	0.65

Table 4: Predictive Model Comparison

1.9 Conclusion

Based on the exploratory analysis, regression models, robustness checks, and nonlinear predictive approaches, the findings can be summarized as follows:

1. **Consistent Positive Association:** Across all modeling approaches, communities with a higher fraction of workers who commute less than 15 minutes tend to exhibit higher levels of upward economic mobility.
2. **Robustness to Geographic Controls:** Including latitude and longitude in the model slightly adjusts the estimates but does not remove the positive association, suggesting the relationship is not purely spatial.
3. **Model Limitations Identified:** Diagnostic checks revealed heteroscedasticity and mild residual dependence, indicating that classical OLS inference may be unreliable without robust standard error adjustments.
4. **Role of Other Contributing Factors:** The dataset does not include important community characteristics such as segregation, school quality, or labor market structure. These unobserved factors may influence both commute patterns and mobility.
5. **Interpretation as Association, Not Causation:** Due to the observational nature of the data and the diagnostic limitations, the results should be interpreted as associational rather than causal. We cannot conclude that short commutes cause higher mobility.
6. **Overall Insight:** Short commutes appear to be a useful indicator of environments where children from low-income families experience greater economic advancement, although further research would be needed to understand the mechanisms underlying this relationship.

2 Problem 2: Air Quality Index Prediction

2.1 Objective

Using high-frequency pollutant and environmental sensor data from Pune’s monitoring stations, the task is to build a robust and scalable model for predicting AQI across diverse locations (e.g., railway stations, bus stops, IT hubs). The challenge includes handling missingness, identifying correlated features and biases, constructing predictive models, and interpreting the learned relationships.

2.2 Dataset Summary and Cleaning

The raw dataset contained 417,637 observations and 31 variables. Substantial missing values were present for humidity, light, CO₂, temperature, sound, and several pollutant concentration fields. We applied:

- **Median imputation** for numeric variables.
- **Mode imputation or rule-based filling** (via AQI thresholds) for categorical fields.
- **Grouping rare categories** in AQI_POLLUTANT into an “Other” class.
- **Datetime parsing** and construction of temporal features (hour, day, month, weekday, weekend).
- **Removal of all-zero columns** and verification that no duplicate rows existed.

After preprocessing, both the training and test sets contained no missing values and shared the same cleaned feature schema.

2.3 Exploratory Data Analysis

2.3.1 AQI Distribution

The AQI distribution is mildly right-skewed, with most readings in the **Moderate** or **Satisfactory** ranges. Extreme high-AQI values (Severe) constitute less than 0.1%, creating heavy class imbalance.

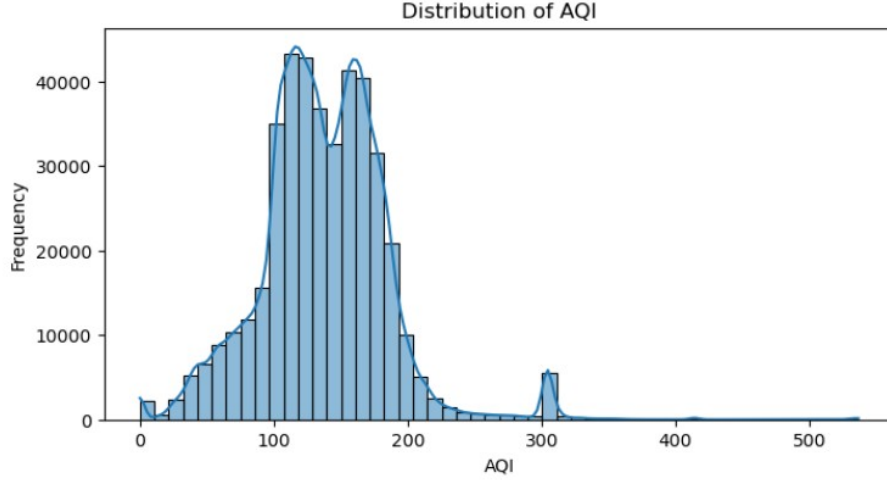


Figure 6: Distribution of AQI Values

2.3.2 AQI Categories

The dataset is dominated by **Moderate (78%)** and **Satisfactory (13%)** categories, with Severe pollution events being rare. This imbalance affects the model’s exposure to high-risk cases.

2.3.3 Pollutant–AQI Relationships

Feature	Correlation with AQI
NO ₂ (max/min)	0.60
Ozone (min/max)	0.21–0.28
CO (max)	0.21
Humidity, Light, Sound	Near 0

Table 5: Key Pollutant Correlations with AQI

NO₂ levels (both max and min) emerged as the strongest drivers of AQI, a finding confirmed throughout our analysis.

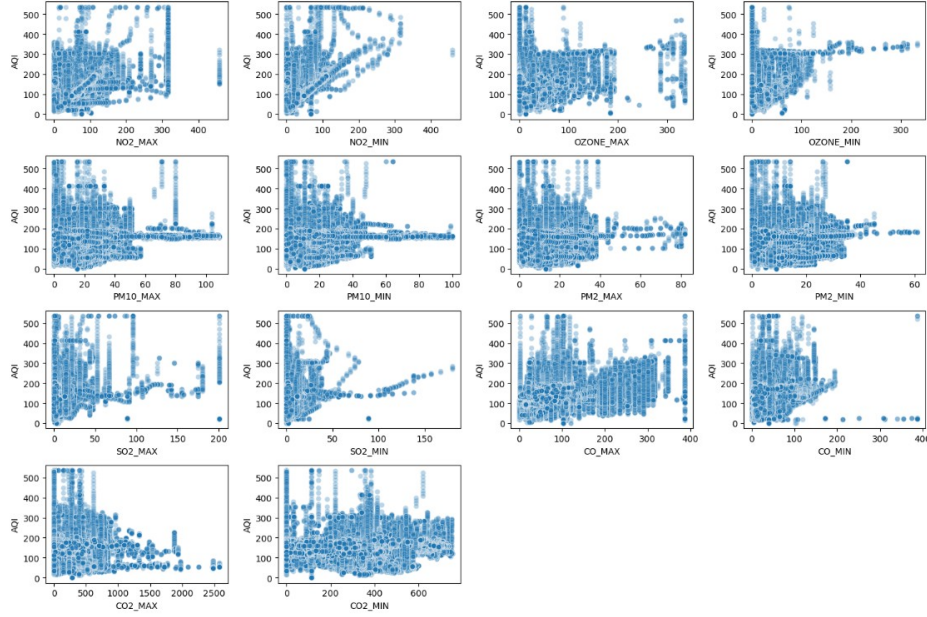


Figure 7: Scatterplots of Major Pollutants vs. AQI (selected features)

2.3.4 Multicollinearity

Several pollutant pairs exhibited strong collinearity:

- PM10_MAX–PM10_MIN (0.95)
- PM2_MAX–PM2_MIN (0.93)
- PM10_MAX–PM2_MAX (0.93)
- NO2_MAX–NO2_MIN (0.87)

This redundancy motivated tree-based models that handle correlated predictors well.

2.3.5 Temporal Bias

Temporal analysis showed that most readings occurred between **April and September**, with sparse data in other months. This constitutes a seasonal sampling bias.

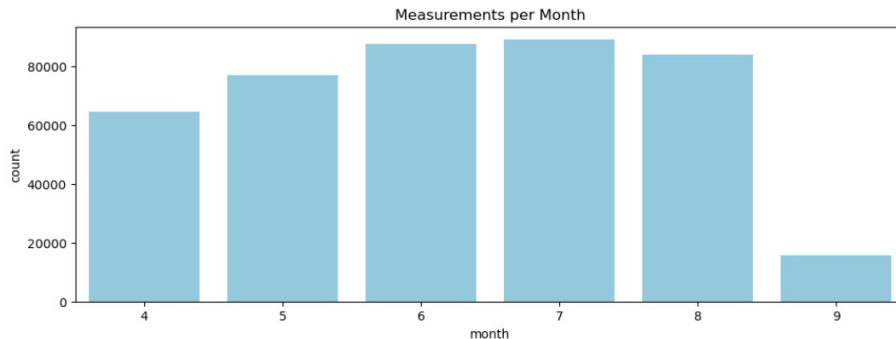


Figure 8: Distribution of Observations by Month

2.3.6 Spatial Structure

A hexbin heatmap of AQI across latitude–longitude showed **no strong spatial pattern**. AQI variability is not strongly location-driven in this dataset; pollutant concentrations and temporal cycles were far more explanatory.

2.4 Model Development

We evaluated three classes of regression models:

- **Linear Regression** (baseline)
- **Random Forest Regressor**
- **XGBoost Regressor**

The training set was split 80–20 into training and validation subsets. The test set contained no AQI values, so RMSE was evaluated on the validation split.

2.5 Model Performance

Model	RMSE	R^2
Linear Regression	26.23	0.702
Random Forest	12.45	0.933
XGBoost	7.68	0.974
XGBoost (5-fold CV)	7.89 ± 0.27	0.973 ± 0.002

Table 6: Model Performance Comparison on Validation Set

Inference: XGBoost offers a dramatic improvement over both linear and tree-ensemble baselines, achieving state-of-the-art accuracy.

2.6 Feature Importance and Interpretability

XGBoost feature importances highlighted:

- **NO₂** (max and min): most influential predictors
- **Ozone** (min/max): strong secondary effects
- **CO_MAX**: meaningful but smaller contribution
- **Temporal features** (hour, month): moderate influence
- **Geographical features**: negligible importance

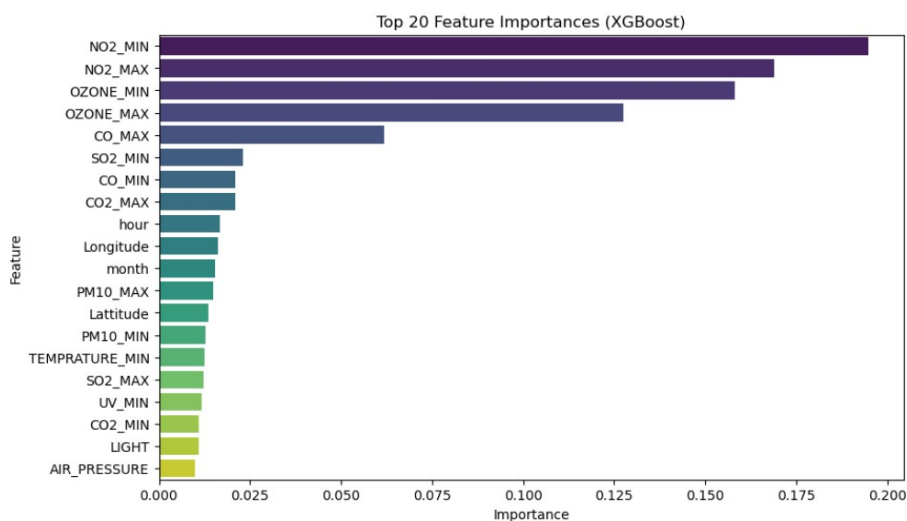


Figure 9: Top 20 Feature Importances (XGBoost)

Partial Dependence Plots (PDPs) confirmed these relationships, revealing smooth increases in AQI with rising NO₂, Ozone, and CO levels.

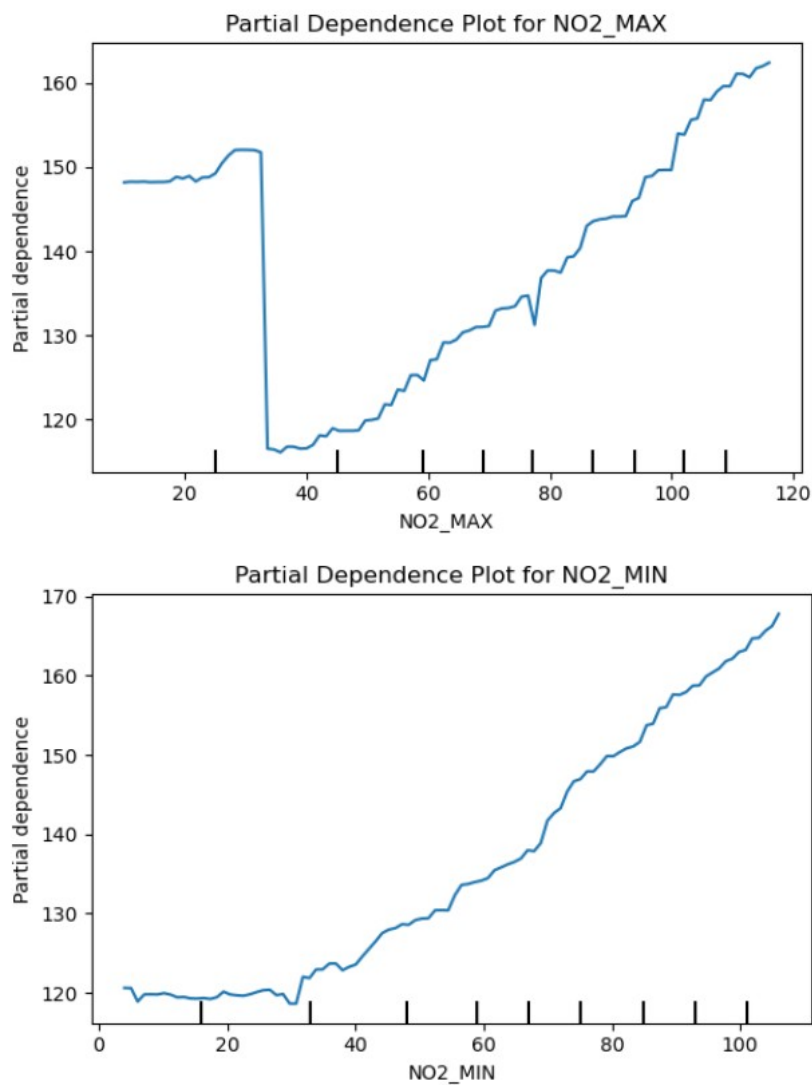


Figure 10: Example PDPs for Key Features

2.7 Deployment and Scalability

The final XGBoost model was saved in JSON format:

- fully portable,
- lightweight,
- reloadable without retraining.

This supports scalable deployment across cities nationwide, as only pollutant measurements are needed for inference.

2.8 Use of Predictions for Automated Actions

AQI predictions can be operationalized in real-time systems to trigger:

- early pollution alerts for public health,
- dynamic traffic-routing interventions,
- industrial emission throttling schedules,
- city-level planning for high-risk zones.

2.9 Conclusion

This problem demonstrated the full lifecycle of building an AQI prediction pipeline: rigorous data cleaning, bias detection, exploratory analysis, model development, interpretability, and deployment. The **XGBoost model achieved RMSE ≈ 7.7 and $R^2 \approx 0.97$** , identifying NO₂, Ozone, and CO as the dominant determinants of AQI. With reliable performance and scalability, the system is well-suited for city-level environmental monitoring and automated decision support.

3 Problem 3: Movie Revenue Prediction

3.1 Objective

The objective of this problem is to build a predictive model for **movie box-office revenue** using a heterogeneous dataset containing metadata, plot summaries, genre information, country and language indicators, and detailed character and actor-level attributes. The task requires:

- performing comprehensive EDA,
- engineering meaningful predictive features,
- fitting machine learning models,
- minimizing RMSE of revenue predictions,
- interpreting model performance and feature importance.

Unlike a recommendation system, this dataset contains **no user-level interactions**, making the task one of **supervised regression**

3.2 Dataset Overview

We use several linked files from the CMU Movie Summary Corpus:

- **movie.metadata** — title, runtime, date, revenue, genres, languages, countries
- **character.metadata** — character names, actor names, gender, height, age
- **plot_summaries** — free-text plot descriptions
- **tvtropes & name clusters** — character-level augmentations

After merging datasets and removing movies with missing revenue, the final modeling set contains **8,401 movies**.

3.3 Exploratory Data Analysis

3.3.1 Distribution of Box-Office Revenue

Revenue is extremely right-skewed, with a small number of films earning disproportionately high amounts.

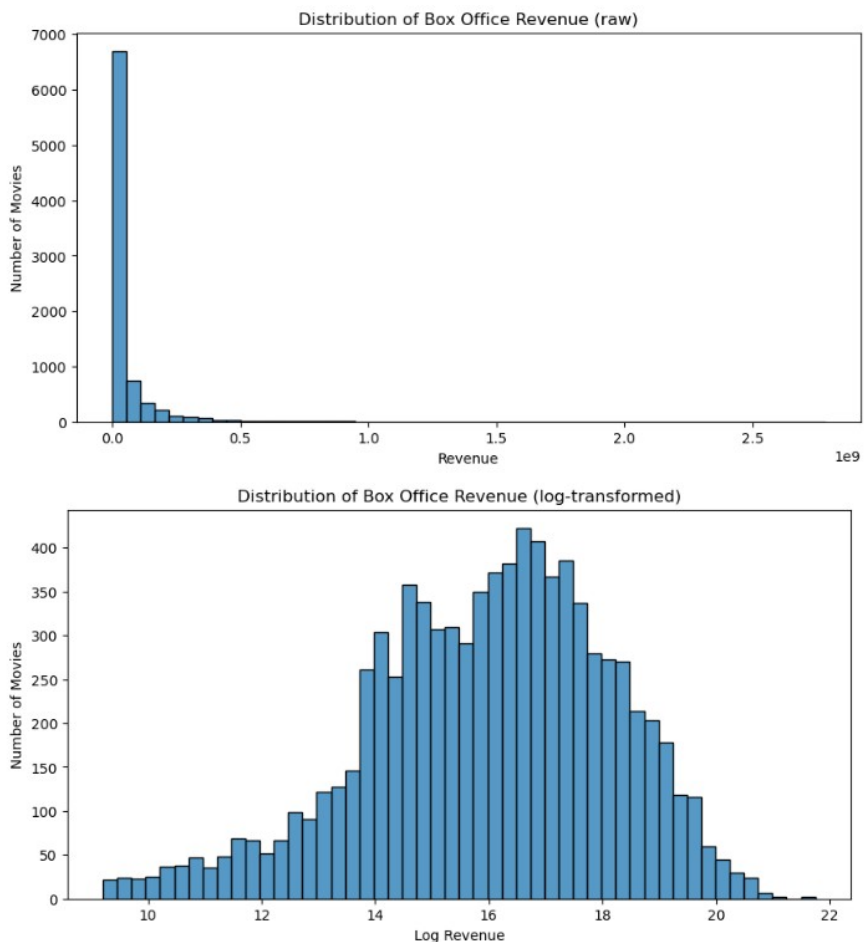


Figure 11: Distribution of Box-Office Revenue (Raw Scale)

A $\log(1 + \text{revenue})$ transform was therefore used for modeling.

3.3.2 Runtime and Release Year Patterns

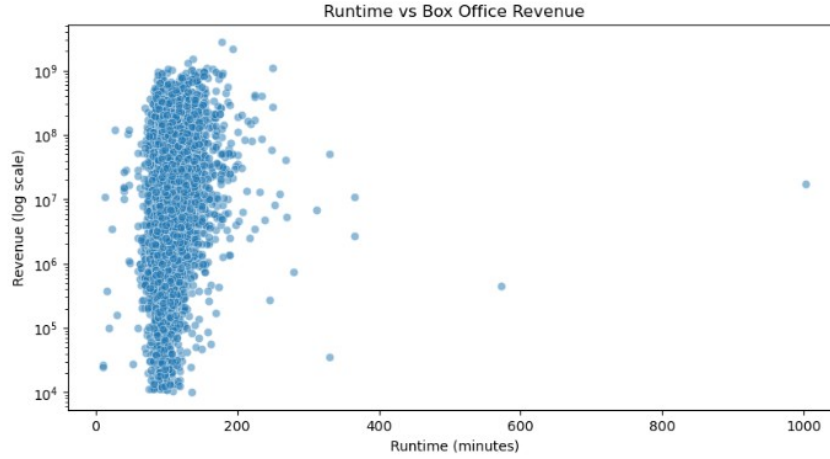


Figure 12: Runtime vs Box-Office Revenue (log scale)

Runtime shows weak correlation with revenue, while year exhibits clustering around 2000–2010 due to data availability.

3.3.3 Plot Length and Revenue

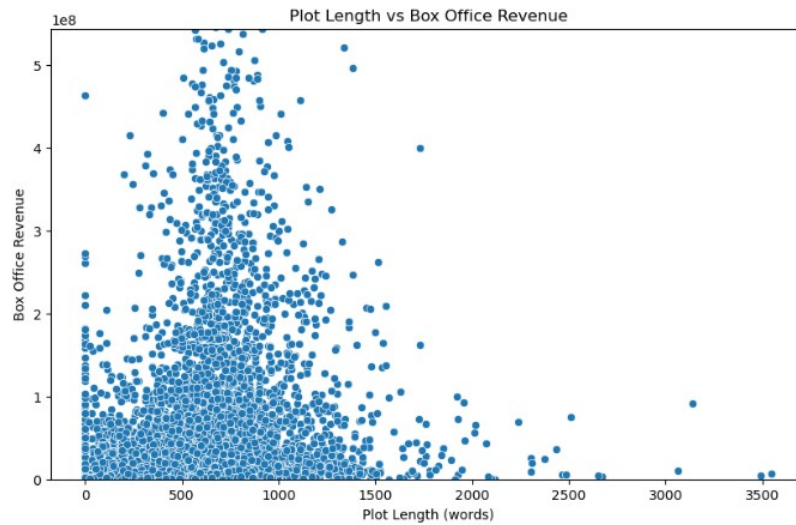


Figure 13: Distribution of Plot Lengths

Most films have plot summaries under 1500 words.

3.3.4 Actor and Character Attributes

Character metadata reveals strong class imbalance:

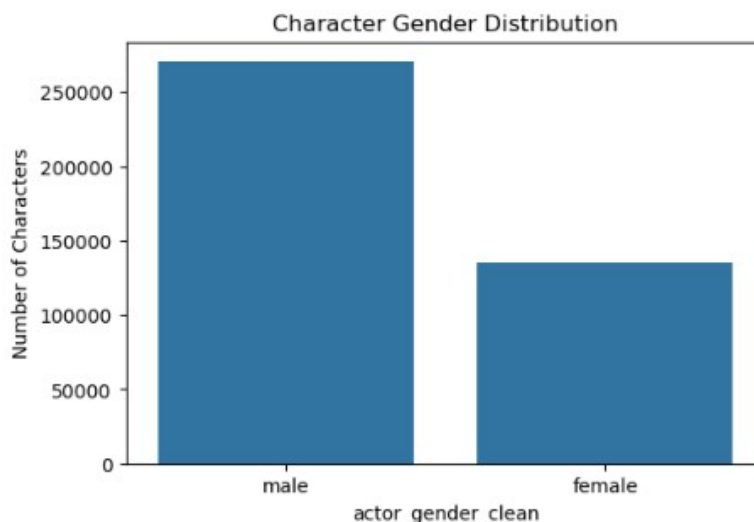


Figure 14: Gender Distribution of Characters

Actor revenue aggregation shows that a small number of actors dominate commercial success.

3.4 Feature Engineering

We engineered four major groups of features:

1. **Numeric Metadata** Runtime, year, number of genres/languages/countries, plot length, number of characters, male/female counts, average actor age and height.
2. **Categorical Metadata** One-hot encodings for genres, languages, and countries, producing several hundred sparse binary indicators.
3. **TF-IDF Plot Features** Plot summaries were transformed using a TF-IDF vectorizer (unigrams + bigrams, 3000 features, min_df=5). These features capture narrative content.
4. **Actor Popularity (Key Feature)** Actor popularity was defined as the **cumulative revenue** of all films an actor appears in. Each movie receives the mean popularity of its cast. This feature proved to be one of the strongest predictors.

All components were combined into a sparse matrix for modeling.

3.5 Models and Methods

We trained several predictive models:

- **Ridge Regression** (TF-IDF only)
- **XGBoost** with multiple feature subsets
- **Random Forest** with metadata and actor popularity

The target variable was modeled in log space:

$$y = \log(1 + \text{revenue})$$

Reported errors include RMSE on both the log and raw revenue scales.

3.6 Model Performance

Model	RMSE(log)	R^2 (log)	RMSE(actual)	R^2 (actual)
Ridge (TF-IDF only)	2.0012	0.1686	99.24M	-0.0048
XGBoost (No Actor)	1.5724	0.4868	82.48M	0.3060
Metadata-only XGB	1.5330	0.5122	76.56M	0.4020
RF + Actor	1.5155	0.5232	66.68M	0.5464
Full XGB (All Features)	1.5059	0.5292	69.63M	0.5053
XGB (Metadata + Actor)	1.4514	0.5627	64.91M	0.5701

Table 7: Model Comparison on Revenue Prediction

The best model is **XGBoost with metadata and actor popularity**, achieving the lowest RMSE and highest R^2 .

3.7 Feature Importance

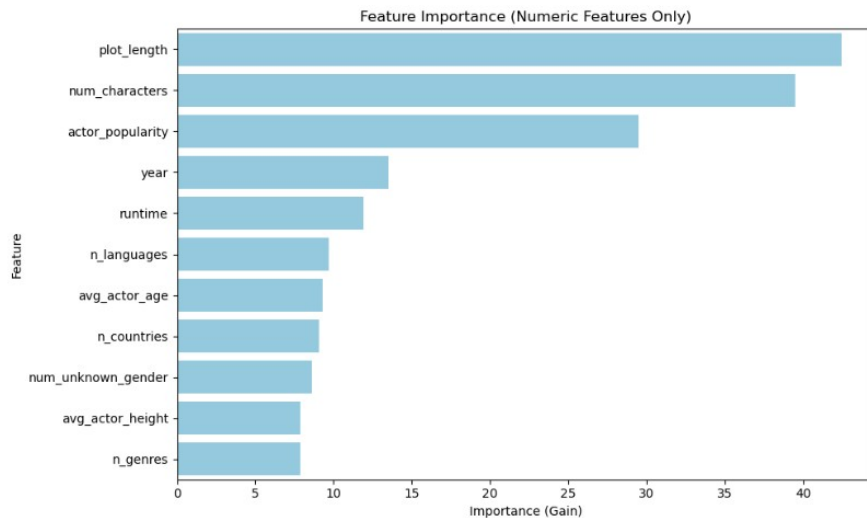


Figure 15: Top Predictors (Metadata + Actor Popularity XGBoost)

Key predictors include:

- plot length
- number of characters
- actor popularity (strongest non-length feature)
- release year
- runtime

Genre one-hot indicators had weak importance, and TF-IDF features offered little signal.

3.8 Conclusion

This problem demonstrates that:

1. **Metadata and actor popularity** contain strong predictive signal for revenue.
2. **Plot text** provides minimal added value; TF-IDF introduces noise.
3. **Actor popularity** is the single feature contributing the largest performance gain, consistent with real-world “star power” effects.
4. **XGBoost outperforms Random Forest and Ridge Regression**, confirming the advantage of boosting on sparse, structured datasets.

While the models achieve an R^2 of approximately 0.57 in log space, revenue forecasting remains fundamentally difficult due to external factors not included in this dataset (e.g., marketing budget, release timing, competition). Thus, richer external data and further feature engineering would be required for industrial-grade forecasting.

Finally, the dataset is well-suited for revenue prediction but needs more data and better feature engineering.